

Data Base Management System (DBMS)

2/09/2026

Name : Harshad Sanjay Teli.

Roll No : 15

Batch : B1

Assignment No 2 – Constraints in SQL

Question 1: Employee Management

Table Structure:

Employee(EmpID, EmpName, Department, Salary, City)

Perform the following SQL operations:

Constraints:

- Create the Employee table with EmpID as the **PRIMARY KEY**.

```
mysql> alter table Employee add primary key(EmpID);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Employee;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| EmpID          | int           | NO   | PRI | NULL    |       |
| EmpName        | varchar(30)   | YES  |     | NULL    |       |
| Department     | varchar(30)   | YES  |     | NULL    |       |
| MonthlySalary  | float(8,2)    | YES  |     | NULL    |       |
| City           | varchar(35)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

b) Apply **NOT NULL** constraint to EmpName and Department

```
mysql> alter table Employee modify EmpName varchar(30) NOT NULL;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(30)	NO		NULL	
Department	varchar(30)	YES		NULL	
MonthlySalary	float(8,2)	YES		NULL	
City	varchar(35)	YES		NULL	

5 rows in set (0.00 sec)

```
mysql> alter table Employee modify Department varchar(30) NOT NULL;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(30)	NO		NULL	
Department	varchar(30)	NO		NULL	
MonthlySalary	float(8,2)	YES		NULL	
City	varchar(35)	YES		NULL	

5 rows in set (0.01 sec)

c) Apply **UNIQUE** constraint to EmpName

```
mysql> alter table Employee modify EmpName varchar(30) Unique;  
Query OK, 0 rows affected (0.08 sec)  
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(30)	YES	UNI	NULL	
Department	varchar(30)	NO		NULL	
MonthlySalary	float(8,2)	YES		NULL	
City	varchar(35)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql>
```

d) Apply **CHECK** constraint to ensure that Salary is greater than 10000.

```
mysql> alter table Employee add check(MonthlySalary>10000);;
Query OK, 4 rows affected (0.05 sec)
Records: 4  Duplicates: 0  Warnings: 0
```

```
ERROR:
No query specified
```

```
mysql> desc Employee;
```

Field	Type	Null	Key	Default	Extra
EmpID	int	NO	PRI	NULL	
EmpName	varchar(30)	YES	UNI	NULL	
Department	varchar(30)	NO		NULL	
MonthlySalary	float(8,2)	YES		NULL	
City	varchar(35)	YES		NULL	

5 rows in set (0.00 sec)

```
mysql> show create table Employee;
```

Table	Create Table
Employee	CREATE TABLE `Employee` (`EmpID` int NOT NULL, `EmpName` varchar(30) DEFAULT NULL, `Department` varchar(30) NOT NULL, `MonthlySalary` float(8,2) DEFAULT NULL, `City` varchar(35) DEFAULT NULL, PRIMARY KEY (`EmpID`), UNIQUE KEY `EmpName` (`EmpName`), CONSTRAINT `Employee_chk_1` CHECK ((`MonthlySalary` > 10000))) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci

1 row in set (0.00 sec)

```
mysql>
```

e) Apply **CHECK** constraint to ensure that Department is one of the following: HR, IT, Finance, Sales.

```
mysql> alter table Employee add CHECK (Department IN ('IT','HR','Sales','Finance'));
Query OK, 1 row affected (0.10 sec)
Records: 1 Duplicates: 0 Warnings: 0

mysql> show create table Employee;
+-----+-----+
| Table | Create Table
+-----+-----+
| Employee | CREATE TABLE `Employee` (
  `EmpID` int NOT NULL,
  `EmpName` varchar(30) DEFAULT NULL,
  `Department` varchar(30) NOT NULL,
  `MonthlySalary` float(8,2) DEFAULT NULL,
  `City` varchar(35) DEFAULT NULL,
  PRIMARY KEY (`EmpID`),
  UNIQUE KEY `EmpName` (`EmpName`),
  CONSTRAINT `Employee_chk_1` CHECK ((`MonthlySalary` > 10000)),
  CONSTRAINT `Employee_chk_2` CHECK ((`Department` in (_utf8mb4'IT',_utf8mb4'HR',_utf8mb4'Sales',_utf8mb4'Finance'))),
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+

1 row in set (0.00 sec)

mysql> 
```

f) Insert **at least 5 employee records** satisfying all the constraints.

```
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(2,'Anurag Lad','HR',45000,'KOP');
Query OK, 1 row affected (0.00 sec)

mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(3,'Vardhan Patki','Sales',30000,'PUNE'),(4,'Rajnandini Patil','Finance',65000,'KOP');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(5,'Shrinath Teli','Marketing',25000,'MUMBAI'),(6,'Avadhut Falle','Testing',75000,'Sangli');
ERROR 3819 (HY000): Check constraint 'Employee_chk_2' is violated.
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(5,'Shrinath Teli','HR',25000,'MUMBAI'),(6,'Avadhut Falle','IT',75000,'Sangli');
Query OK, 2 rows affected (0.01 sec)
Records: 2 Duplicates: 0 Warnings: 0

mysql> selcet * from Employee;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'selcet * from Employee;
line 1
mysql> selectet * from Employee;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'selectet
line 1
mysql> select * from Employee;
+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+
| 1 | Harshad Teli | IT         | 40000.00      | Kolhapur |
| 2 | Anurag Lad   | HR         | 45000.00      | KOP      |
| 3 | Vardhan Patki | Sales      | 30000.00      | PUNE     |
| 4 | Rajnandini Patil | Finance   | 65000.00      | KOP      |
| 5 | Shrinath Teli | HR         | 25000.00      | MUMBAI   |
| 6 | Avadhut Falle | IT         | 75000.00      | Sangli   |
+-----+-----+
6 rows in set (0.00 sec)
```

g) Display all employee records.

```
mysql> select * from Employee;
```

EmpID	EmpName	Department	MonthlySalary	City
1	Harshad Teli	IT	40000.00	Kolhapur
2	Anurag Lad	HR	45000.00	KOP
3	Vardhan Patki	Sales	30000.00	PUNE
4	Rajnandini Patil	Finance	65000.00	KOP
5	Shrinath Teli	HR	25000.00	MUMBAI
6	Avadhut Falle	IT	75000.00	Sangli

```
6 rows in set (0.00 sec)
```

h) Try to insert an employee record with a duplicate EmpID and observe the result.

```
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(1,'Nilam Patil','HR',55000,'KOP');
ERROR 1062 (23000): Duplicate entry '1' for key 'Employee.PRIMARY'
mysql>
```

i) Try to insert an employee record with a duplicate EmpName and observe the result..

```
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(7,'Harshad Teli','IT',35000,'KOP');
ERROR 1062 (23000): Duplicate entry 'Harshad Teli' for key 'Employee.EmpName'
mysql>
```

j) Try to insert an employee record with Salary less than or equal to 10000 and observe the result.

```
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)values(7,'Sanika Patil','IT',8000,'SATARA');
ERROR 3819 (HY000): Check constraint 'Employee_chk_1' is violated.
mysql>
```

k) Try to insert an employee record without providing EmpName and observe the result.

```
mysql> insert into Employee(EmpID,EmpName,Department,MonthlySalary,City)value(45,'Testing',24000,'KOP');
ERROR 1136 (21S01): Column count doesn't match value count at row 1
mysql>
```

1) Display the table structure and verify the constraints applied.

```
mysql> show create table Employee;
+-----+-----+
| Table      | Create Table
+-----+-----+
| Employee   | CREATE TABLE `Employee` (
  `EmpID` int NOT NULL,
  `EmpName` varchar(30) DEFAULT NULL,
  `Department` varchar(30) NOT NULL,
  `MonthlySalary` float(8,2) DEFAULT NULL,
  `City` varchar(35) DEFAULT NULL,
  PRIMARY KEY (`EmpID`),
  UNIQUE KEY `EmpName` (`EmpName`),
  CONSTRAINT `Employee_chk_1` CHECK ((`MonthlySalary` > 10000))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

Question 2: Student Management

Table Structure:

Student(StudentID, StudentName, Course, Age, Email)

Perform the following SQL operations:

Constraints:

a) Create the Student table with StudentID as the **PRIMARY KEY**.

```
mysql> alter table Student add PRIMARY KEY(StudentID);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | NO   | PRI | NULL    |       |
| StudentName    | varchar(40)   | YES  |     | NULL    |       |
| Course         | varchar(30)   | YES  |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | varchar(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.01 sec)

mysql> 
```

b) Apply **NOT NULL** constraint to StudentName and Course.

```
mysql> alter table Student modify StudentName varchar(40) NOT NULL;
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> alter table Student modify Course varchar(30)  NOT NULL;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | NO   | PRI | NULL    |       |
| StudentName    | varchar(40)   | NO   |     | NULL    |       |
| Course         | varchar(30)   | NO   |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | varchar(10)   | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> 
```

c) Apply **UNIQUE** constraint to Email.

```
mysql> alter table Student modify Email varchar(20) UNIQUE;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Studentl;
ERROR 1146 (42S02): Table 'Harshad.Studentl' doesn't exist
mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | NO   | PRI | NULL    |       |
| StudentName    | varchar(40)   | NO   |     | NULL    |       |
| Course         | varchar(30)   | NO   |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | varchar(10)   | YES  |     | NULL    |       |
| Email          | varchar(20)   | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> 
```

d) Apply **CHECK** constraint to ensure that Age is greater than or equal to 18.

```
mysql> alter table Student add check (StudentAge >= 18);
Query OK, 4 rows affected (0.08 sec)
Records: 4 Duplicates: 0 Warnings: 0

mysql> show create table Student;
+-----+-----+
| Table | Create Table |
+-----+-----+
| Student | CREATE TABLE `Student` (
  `StudentID` int NOT NULL,
  `StudentName` varchar(40) NOT NULL,
  `Course` varchar(30) NOT NULL,
  `StudentAge` int DEFAULT NULL,
  `MobileNo` varchar(10) CHARACTER SET utf8mb3 COLLATE utf8mb3_general_ci DEFAULT NULL,
  `Email` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`StudentID`),
  UNIQUE KEY `Email` (`Email`),
  CONSTRAINT `Student_chk_1` CHECK ((`StudentAge` >= 18))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)

mysql> 
```

e) Apply **CHECK** constraint to ensure that Course is one of the following: MCA, MBA, BCA, MSc.

```
mysql> alter table Student add CHECK (Course IN ('MBA','MCA','Msc','BCA'));
Query OK, 4 rows affected (0.07 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql> show create table Student;
+-----+-----+
| Table | Create Table
+-----+-----+
| Student | CREATE TABLE `Student` (
  `StudentID` int NOT NULL,
  `StudentName` varchar(40) NOT NULL,
  `Course` varchar(30) NOT NULL,
  `StudentAge` int DEFAULT NULL,
  `MobileNo` varchar(10) CHARACTER SET utf8mb3 COLLATE utf8mb3_general_ci DEFAULT NULL,
  `Email` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`StudentID`),
  UNIQUE KEY `Email` (`Email`),
  CONSTRAINT `Student_chk_1` CHECK ((`StudentAge` >= 18)),
  CONSTRAINT `Student_chk_2` CHECK ((`Course` in (_utf8mb4'MBA',_utf8mb4'MCA',_utf8mb4'Msc'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

f) Insert **at least 5 student records** satisfying all the constraints.

(From old Data in the Student Table will be Display)

```
mysql> select * from Student;
+-----+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo | Email |
+-----+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | MBA | 18 | 9970898012 | harshad@gmail.com |
| 2 | Anurag Patil | MCA | 22 | 9156371425 | anurag@gmail.com |
| 4 | Rajveer Patil | BCA | 21 | 7894561245 | rajveer@gmail.com |
| 5 | Shruti Patili | MSc | 23 | 9970898012 | shruti@gmail.com |
| 6 | Anjali Patil | BCA | 21 | 9970898012 | anjali@gmail.com |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

g) Display all student records.

```
mysql> select * from Student;
+-----+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo | Email |
+-----+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | MBA | 18 | 9970898012 | harshad@gmail.com |
| 2 | Anurag Patil | MCA | 22 | 9156371425 | anurag@gmail.com |
| 4 | Rajveer Patil | BCA | 21 | 7894561245 | rajveer@gmail.com |
| 5 | Shruti Patili | MSc | 23 | 9970898012 | shruti@gmail.com |
| 6 | Anjali Patil | BCA | 21 | 9970898012 | anjali@gmail.com |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

h) Try to insert a student record with a duplicate StudentID and observe the result.

```
mysql> INSERT INTO Student(StudentID,StudentName,Course,StudentAge,MobileNo,Email) VALUES (7, 'Kartika', 'BCA', 19, '7029525647', 'kartika@gmail.com');
ERROR 1062 (23000): Duplicate entry '7' for key 'Student.PRIMARY'
mysql>
```

i) Try to insert a student record with a duplicate Email and observe the result.

```
mysql> INSERT INTO Student(StudentID,StudentName,Course,StudentAge,MobileNo,Email) VALUES (8, 'Kartika', 'BCA', 19, '7029525647', 'anjali@gmail.com');
ERROR 1062 (23000): Duplicate entry 'anjali@gmail.com' for key 'Student.Email'
mysql>
```

j) Try to insert a student record with Age less than 18 and observe the result.

```
mysql> INSERT INTO Student(StudentID, StudentName, Course, StudentAge, MobileNo, Email) VALUES (10, 'Aarav Joshi', 'BCA', 16, '9876543210', 'aarav.j@gmail.com');
ERROR 3819 (HY000): Check constraint 'Student_chk_1' is violated.
mysql>
```

k) Try to insert a student record without providing StudentName and observe the result.

```
mysql> INSERT INTO Student(StudentID, Course, StudentAge, MobileNo, Email)
-> VALUES (9, 'MCA', 20, '9876543210', 'test9@gmail.com');
ERROR 1364 (HY000): Field 'StudentName' doesn't have a default value
mysql>
```

1) Display the table structure and verify the constraints applied.

```
mysql> show create table Student;
+-----+-----+
| Table | Create Table
+-----+-----+
| Student | CREATE TABLE `Student` (
  `StudentID` int NOT NULL,
  `StudentName` varchar(40) NOT NULL,
  `Course` varchar(30) NOT NULL,
  `StudentAge` int DEFAULT NULL,
  `MobileNo` varchar(10) CHARACTER SET utf8mb3 COLLATE utf8mb3_general_ci DEFAULT NULL,
  `Email` varchar(20) DEFAULT NULL,
  PRIMARY KEY (`StudentID`),
  UNIQUE KEY `Email` (`Email`),
  CONSTRAINT `Student_chk_1` CHECK ((`StudentAge` >= 18)),
  CONSTRAINT `Student_chk_2` CHECK ((`Course` in (_utf8mb4'MBA',_utf8mb4'MCA',_utf8mb4'Msc',_utf8mb4'BCA')))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci |
+-----+-----+
1 row in set (0.00 sec)

mysql> 
```

c) Add a new column MobileNo.

```
mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(50)	YES		NULL	
Course	varchar(50)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.00 sec)

d) Modify the size of the Course column.

```
mysql> alter table Student modify Course varchar(30);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(50)	YES		NULL	
Course	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.00 sec)

e) Modify the size of the StudentName column.

```
mysql> alter table Student modify StudentName varchar(40);;
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

ERROR:
No query specified

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(40)	YES		NULL	
Course	varchar(30)	YES		NULL	
Age	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.01 sec)

f) Rename the column Age to StudentAge.

```
mysql> alter table Student change Age StudentAge int;
Query OK, 0 rows affected (0.01 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> desc Student;
```

Field	Type	Null	Key	Default	Extra
StudentID	int	YES		NULL	
StudentName	varchar(40)	YES		NULL	
Course	varchar(30)	YES		NULL	
StudentAge	int	YES		NULL	
Email	varchar(20)	YES		NULL	
MobileNo	int	YES		NULL	

6 rows in set (0.00 sec)

g) Drop the Email column

```
mysql> alter table Student drop email;
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc Student;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | YES  |     | NULL    |       |
| StudentName    | varchar(40)   | YES  |     | NULL    |       |
| Course         | varchar(30)   | YES  |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

h) Rename the table Student to StudentMaster.

```
mysql> rename table Student to StudentMaster;
Query OK, 0 rows affected (0.02 sec)

mysql> desc StudentMaster;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| StudentID      | int           | YES  |     | NULL    |       |
| StudentName    | varchar(40)   | YES  |     | NULL    |       |
| Course         | varchar(30)   | YES  |     | NULL    |       |
| StudentAge     | int           | YES  |     | NULL    |       |
| MobileNo       | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

DML Commands:

i) Insert **at least 5 student records** into the table.

```
mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)value(1,'Harshad Teli','BCS',18,9970898012);
Query OK, 1 row affected (0.01 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(2,'Anurag Patil','BCA',18,9156371425);
Query OK, 1 row affected (0.02 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(3,'Vardhan Patki','BCOM',20,7029536728);
Query OK, 1 row affected (0.02 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(4,'Rajveer Patil','BA',21,7029536728);
Query OK, 1 row affected (0.01 sec)

mysql> insert into StudentMaster(StudentID,StudentName,Course,StudentAge,MobileNo)values(5,'Shruti Patili','BSC',23,9970898012);
Query OK, 1 row affected (0.03 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7029536728 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

j) Display all records from the table.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

k) Update the Course of a student using StudentID.

```
mysql> update StudentMaster set Course = 'B.Tech' where StudentID=4;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

l) Update the StudentAge of a student using StudentID.

```
mysql> update StudentMaster set StudentAge = 22 where StudentID = 2;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

m) Update the MobileNo of a student using StudentID.

```
mysql> update StudentMaster set MobileNo= 7894561245 where StudentID=4;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 18 | 9156371425 |
| 3 | Vardhan Patki | BCOM | 20 | 7029536728 |
| 4 | Rajveer Patil | BA | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

n) Delete one student record using StudentID.

```
mysql> delete from StudentMaster where StudentID = 3;
Query OK, 1 row affected (0.02 sec)

mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

o) Display the records after performing the update and delete operations.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | MobileNo |
+-----+-----+-----+-----+-----+
| 1 | Harshad Teli | BCS | 18 | 9970898012 |
| 2 | Anurag Patil | BCA | 22 | 9156371425 |
| 4 | Rajveer Patil | B.Tech | 21 | 7894561245 |
| 5 | Shruti Patili | BSC | 23 | 9970898012 |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```